MIL-G-46015B(ME)
28 March 1983
SUPERSEDING
MIL-G-46015A(MR)
18 March 1976

MILITARY SPECIFICATION

GASOLINE, AUTOMOTIVE, COMBAT, REFEREE GRADE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two grades of gasoline for use in research, development and proof-testing of all gasoline engines (except aircraft engines), gasoline-powered auxiliary units, and other supply items designed to handle or operate on tactical grades of motor gasoline. The grades are provided to meet different operating and environmental conditions, with their intended uses described in 6.1.

1.2 Classification. The gasolines shall be of the following grades (see 6.1):

Grade I - Engine endurance gasoline (all purpose grade).
Grade II - Cold start gasoline - for temperatures below 0° C (low-temperature, all purpose grade).

1.3 Definition. Referee grade gasoline is defined as gasoline representing the minimal or marginal quality level which can be procured under specification MIL-G-3056 while meeting all its requirements. It is designed to be equivalent to the quality of OCONUS gasoline production or that available for production in times of a national emergency. It is used for research, development and proof testing to assure that equipment will perform adequately with all gasolines procured under MIL-G-3056.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9130

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2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-I-25017	- Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble (Metric).

STANDARDS

FEDERAL

FED-STD-791	- Lubricants, Liquid Fuels, and Related Products, Methods of Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-290	- Packaging of Petroleum and Related Products.

QUALIFIED PRODUCTS LIST

QPL-25017	- Inhibitor, Corrosion, Fuel Soluble.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS

Title 49, Part 100-199, Department of Transportation (DOT) Rules and Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D86 - Distillation of Petroleum Products.
D130 - Detection of Copper Corrosion from Petroleum Products by the Copper Strip Tarnish Test.
D323 - Vapor Pressure of Petroleum Products (Reid Method).
D381 - Existent Gum in Fuels by Jet Evaporation.
D439 - Specification for Automotive Gasoline.
D525 - Oxidation Stability of Gasoline (Induction Period Method).
D1266 - Sulfur in Petroleum Products (Lamp Method).
D1298 - Density, Relative Density, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method.
D1319 - Test for Hydrocarbon Types in Liquid Petroleum Products by Fluorescent Indicator Adsorption.
D2276 - Particulate Contaminant in Aviation Turbine Fuels.
D2533 - Vapor-Liquid Ratio of Gasoline.
D2547 - Lead in Gasoline, Volumetric Chromate Method.
D2551 - Vapor Pressure of Petroleum Products (Micromethod).
D2599 - Lead in Gasoline by X-Ray Spectrometry.
D2622 - Sulfur in Petroleum Products (X-Ray Spectrographic Method).
D2699 - Knock Characteristics of Motor Fuels by the Research Method.
D2700 - Knock Characteristics of Motor and Aviation-Type Fuels by the Motor Method.
D2885 - Research and Motor Method Octane Ratings Using On-Line Analyzers.
D4057 - Sampling of Petroleum and Petroleum Products.
D4176 - Free Water and Particulate Contamination in Distillate Fuels (Clear and Bright Pass/Fail Procedures).

(The test methods listed above are included in parts 23, 24, 25, and 47 of the Annual Book of ASTM Standards or in the ASTM Manual for Rating Fuels by Motor and Research Methods. The methods can also be purchased separately. Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. The gasolines shall be volatile hydrocarbon fuels.

3.2 Additives. Additives and additive concentrations for both type I and type II gasolines shall be as specified in 3.2.1 through 3.2.5.

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3.2.1 Oxidation inhibitors. The gasolines shall contain not less than 28 grams nor more than 43 grams of oxidation inhibitors (active ingredient) per cubic metre of gasoline. Any one or a combination of the following oxidation inhibitors may be used:

a. N,N'-di-sec-butyl-p-phenylenediamine
b. N,N'-diisopropyl-p-phenylenediamine
c. N,N'-dioctyl-p-phenylenediamine
d. N,N'-bis(1,4-dimethylpentyl)-p-phenylenediamine
e. N,N'-di-sec-butyl-o-phenylenediamine
f. 2,6-di-tert-butylphenol
g. 2,6-di-tert-butyl-4-methylphenol
h. 2,4-dimethyl-6-tert-butylphenol
i. Triethylenetetramine di(monononylphenolate)
j. Mixed tert-butyl phenols
k. N-sec-butyl,N',-phenyl-o-phenylenediamine
l. Mixed 2,6-dialkyl and 2,4,6,-trialkyl phenols (containing mixed hexyl and heptyl groups)
m. 2,4-di-tert-butylphenol (60 weight percent minimum) and mixed tert-butyl phenols (40 weight percent maximum)
n. 2,4-di-tert-butylphenol (containing mono- and tri-tert-butyl phenols)
o. Mixed di- and triisopropyl phenols (75 weight percent minimum) and mixed di- and tri-tert-butyl phenols (25 weight percent maximum)

3.2.2 Metal deactivators. The gasolines shall contain not less than 2.8 grams nor more than 8.6 grams of an approved metal deactivator (active ingredient) per cubic metre of gasoline. Any one of the following metal deactivators may be added separately or in combination with an approved oxidation inhibitor.

a. N,N'-disalicylidene-1,2-ethanediamine
b. N,N'-disalicylidene-1,2-propanediamine
c. N,N'-disalicylidene-1,2-cyclohexanediamine
d. Disalicylidene-N-methyl-dipropylenetriamine

3.2.3 Corrosion inhibitors. An approved corrosion inhibitor conforming to MIL-I-25017 shall be added to the gasolines. The quantity of corrosion inhibitor added shall not be less than 9.0 grams per cubic metre nor more than the maximum allowable concentration listed in QPL-25017 for the specific corrosion inhibitor used.

3.2.4 Antiknock compounds. Antiknock compounds shall be of a lead alkyl type, identified as tetraethyllead, methyltriethyllead, dimethyldiethyllead, trimethylethyllead, tetramethyllead, or admixtures thereof. The lead alkyl compound mixture shall contain a stoichiometric quantity of organic bromine or chlorine scavenger compounds, or admixtures thereof, and the necessary solvents and dye. The content of lead antiknock compounds calculated as metallic lead shall not be less than 0.13 grams or more than 0.65 grams per litre.

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3.2.5 Other additives. The gasolines shall not contain other additives such as detergents, solvent oils, dispersants, etc., except dyes which are required to meet the color requirement.

3.3 Physical and chemical requirements. The finished gasolines shall be as specified in table I. Where the term "Report" appears in the requirements of table I, it means the value of that property is not limited, but shall be determined by the contractor and reported to the acquiring agency.

TABLE I. Physical and chemical requirements.

Property	Grade I	Grade II
Distillation		
Initial boiling point	Report	Report
10% evap., °C	50 to 60	45 to 55
50% evap., °C	88 to 105	85 to 95
90% evap., °C	160 to 180	136 to 150
End point, °C	205 max.	Report
Residue, % vol, max.	2	2
Temperature, °C, min., at V/L ratio = 20 1/	56	41
Reid vapor pressure:		
kPa 38° C	48 to 62	83 to 97
Unwashed gum, mg/100 mL, max.	4	4
Lead, grams/litre	0.13 to 0.65	0.13 to 0.65
Sulfur, % wt 2/	0.08 to 0.12	0.10 max.
Corrosiveness at 50° C, ASTM No., max.	1	1
Oxidation stability, minutes, min.	480	480
Particulate contaminant, mg/L, max.	1.0	1.0
Composition, % vol.		
Saturates	Report	Report
Olefins	Report	Report
Aromatics	25 to 45	Report
Color	Red, equal to standard	Red, equal to standard
Density, Kg/L at 15° C	Report	Report
Octane number, research method	89 to 91	89 to 91
Octane number, motor method	81 to 83	81 to 83
Free water and particulate contamination	Pass	Pass

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1/ At 101.3 kPa pressure.

2/ Sulfur should be naturally-occuring. If addtional sulfur is required, only tertiary butyl disulfide shall be added.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot.

4.2.1 Bulk lot. An indefinite quantity of a homogeneous blend of one type of gasoline offered for acceptance in a single, isolated container; or manufactured in a single plant run (not exceeding 24 hours), through the same processing equipment, with no change in the ingredient materials.

4.2.2 Packaged lot. An indefinite number of 55-gallon drums or smaller unit containers of identical size and type, offered for acceptance, and filled with a homogeneous blend of one type of gasoline from a single, isolated container; or filled with a homogenous blend of one type of gasoline manufactured in a single plant run (not exceeding 24 hours), through the same processing equipment, with no change in the ingredient materials.

4.3 Sampling.

4.3.1 Sampling for inspection of filled containers. Take a random sample of filled containers from each lot in accordance with MIL-STD-105, at inspection level II. Acceptable quality level (AQL) shall be 2.5 percent defective.

4.3.2 Sampling for tests. Take samples for tests in accordance with ASTM method D4057.

4.4 Inspection. Perform inspection in accordance with method 9601 of FED-STD-791.

4.4.1 Examination of filled containers. Examine samples selected in accordance with 4.3.1 for compliance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and marking requirements. Reject any container having one or more defects or under the required fill. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, reject the lot represented by the sample.

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4.5 Classification of tests. All tests are quality conformance tests.

4.6 Test procedures. Perform tests in accordance with table II.

TABLE II. Test methods.

Test	ASTM method number
Distillation	D86
Vapor-liquid (V/L) ratio	D2533 1/
Reid vapor pressure	D323 or D2551 2/
Unwashed gum	D381 3/
Lead	D2547 or D2599
Sulfur	D1266 or D2622
Corrosiveness at 50° C	D130
Oxidation stability	D525
Particulate contaminant	D2276 4/
Composition	D1319
Color	5/
Density	D1298
Octane rating, research method	D2699 or D2885
Octane rating, motor method	D2700 or D2885
Free water and particulate contamination	D4176

1/ As an alternative, calculate the temperture at a vapor-liquid ratio of 20:1 in accordance with ASTM D439, paragraph X1.3. However, when calculated values are questionable, use ASTM D2533 as the referee V/L method.

2/ ASTM D2551, Vapor Pressure of Petroleum Products (Micromethod) may be used as an alternate method for determining vapor pressure.

3/ Unwashed gum is defined in ASTM D381, paragraph 5.2. Follow the procedure for unwashed gum given in D381.

4/ Method A, total contaminant.

5/ Make assessments of color in accordance with FED-STD-791, Method 103.

4.7 Report of analysis. The contractor shall provide copies of analysis reports, giving the results of these tests with a statement of the type and concentration of the additives used in the gasoline.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified (see 6.2), packaging, packing, and marking shall be in accordance with MIL-STD-290.

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5.2 Special marking for packaging lots. Each container shall be labeled in accordance with CFR Title 49, Part 172; also each container in a packaged lot shall bear the following cautionary label:

WARNING: CONTAINS LEADED GASOLINE
Do not swallow.
Avoid breathing vapor.
Avoid contact with skin.

6. NOTES

6.1 Intended use. Gasolines covered by this specification are intended for use in research, development and proof-testing or qualification of spark-ignition and ground-based turbine engines, components, vehicle heaters, etc. The use of referee fuel in conjunction with Development Testing (DTI through DTIII) is considered necessary to uncover potential operational problems with respect to performance and to permit data correlation of all testing sites regardless of geographical location. Referee fuels should not be used in engines or other equipment in the field as a substitute for MIL-G-3056 gasoline without approval of the engine manufacturer or of the U.S. Army Mobility Equipment Research and Development Command, ATTN: DRDME-GL, Fort Belvoir, VA 22060.

6.1.1 Grade I - Engine endurance gasoline. This gasoline is intended for use in all engine and vehicle qualification, proof, engineering or pilot model testing in conjuction with appropriate reference engine oils to assure compatibility with military standard supply items.

6.1.2 Grade II - Cold start gasoline. This gasoline is intended for use in cold-room studies to permit evaluation of cold starting and warm-up operational characteristics.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type of gasoline required (see 1.2 and 6.1).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. Quantity of gasoline required. The unit of purchase is one U.S. gallon (3.785 litres at 15° C).
e. Type and size of containers required (see 5.1).
f. Level of packaging and level of packing required (see 5.1).

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Review activities:
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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